

Business Understanding Report: Heart Failure Prediction

Business Problem

Heart failure occurs when the heart cannot pump enough blood to meet the body's needs. It's usually triggered by other health conditions affecting the heart muscles, including coronary artery disease, heart attack, high blood pressure, heart valve disorders, congenital defects, cardiomyopathy, myocarditis, diabetes, and pulmonary hypertension.

While heart failure is often caused by underlying health conditions, each individual's health risks are unique. Many factors like age, family history, and lifestyle can contribute, but early detection with advanced diagnostic tools can help manage and reduce the likelihood of serious damage.

Therefore, a heart failure prediction model would be effective in detecting the likelihood or the presence of heart failure in individuals.

Project Objective

The objective of this project is to build a machine learning model that predicts whether a patient is likely to develop heart failure using clinical features.

Target variable

The target variable is the outcome indicating the presence or absence of heart failure.

Importance of predictive modeling

Early detection and risk stratification:

Heart failure often develops gradually, with early signs (fatigue, mild fluid retention, changes in exercise tolerance) that are easy to miss in routine care. Predictive models — using structured data like EHR history, lab values, vitals, and comorbidities (diabetes, hypertension) — can flag patients at elevated risk before a clinical diagnosis is obvious, allowing earlier intervention.

Readmission and mortality prediction:

A large share of published clinical ML work in this space focuses on predicting 30-day readmission or short-term mortality after a heart failure diagnosis or hospitalization. This is useful because readmissions are costly and often preventable with better discharge planning, closer follow-up, or medication adjustments.

Resource allocation in constrained health systems:

This is probably the angle most relevant to your interests — in low-resource settings, predictive triage tools can help prioritize limited specialist time, diagnostic equipment (echocardiography, etc.), or ICU beds toward the patients most likely to deteriorate.

Expected Business Impact

A heart failure prediction model can create value by enabling earlier intervention, which allows treatment adjustments while the patient is still stable rather than after a crisis.

It can also support personalized care pathways by helping healthcare providers assign different follow-up levels based on risk. In addition, earlier detection may reduce emergency hospitalizations, lower mortality risk, and improve overall resource use in healthcare settings.